Diagnostic Analysis of Neural Representations on the IndiVision Traditional Clothing Dataset

Deep Learning for Computer Vision (CSE3292) Phase 2 Submission

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Abstract

This report details the mechanistic functioning and learning behavior of deep neural networks across three distinct learning paradigms on the IndiVision dataset, a custom curation of 3,218 Indian traditional clothing images. Moving beyond black-box accuracy metrics, we performed diagnostic experiments to understand why certain architectures succeed or fail. Our findings indicate that Receptive Field size strictly dictates a Convolutional Neural Network's ability to classify texture vs. structural shapes. Furthermore, Vision Transformer (ViT) ablation studies highlight immense intra-layer redundancy and head-specific spatial responsibilities. Finally, our layer freezing experiments establish that preserving early-layer ImageNet weights (Strategy B) serves as vital regularization against overfitting for small, culturally diverse datasets, yielding an optimal accuracy of 83.02%.

1 Introduction

Modern Computer Vision research frequently treats foundation models as opaque tools optimized purely for benchmark performance. In Phase 1 of this project, we pioneered the *IndiVision Dataset*, encompassing 12 distinct classes of Indian traditional wear. The dataset intrinsically blends high-frequency micro-textures (e.g., Bandhani prints) with macro-structural dependencies (e.g., the global drape of a Saree or Sherwani).

In Phase 2, we pivot to a diagnostic approach—treating deep neural networks as subjects of scientific inquiry. Guided by CO-4 requirements, we conducted targeted experiments across Convolutional Neural Networks (CNNs), Vision Transformers (ViTs), and Transfer Learning regimes. Our methodology focuses heavily on ablations, layer freezing, and intermediate representation measurements to dissect network behavior.

2 Unit I: Receptive Field Architecture in CNNs

2.1 Motivation and Hypothesis

Different clothing categories mandate specialized localized logic. We hypothesized that shallow CNNs (with small effective receptive fields) would inherently capture isolated, repeating local patches of patterns, thereby succeeding on texture-dominant classes. Deep CNNs (with large fields), conversely, would excel at recognizing global silhouettes but aggressively discard structural noise—hurting fine-grained fabric recognition.

2.2 Methodology

We executed Linear Probing against three frozen ResNet variants: ResNet-18 (Shallow), ResNet-50 (Mid), and ResNet-152 (Deep), ensuring standard ImageNet-1K V1 initial weights. By freezing the backbone and training solely a minimal classification head, we forced the exact intermediate convolution layers to expose their unaltered feature representation abilities.

2.3 Findings and Analysis

The experimental results (Table 1) stringently align with the receptive field hypothesis. Structural clothing heavily relies on deep, global contexts:

- **Sherwani:** Achieved 96.2% on Res-18 and flawlessly scaled to 100% and 98.1% on deeper architectures.

- **Saree:** Showed excellent deep-network coherence (93.3% on Res-152).

Conversely, heavily textured fabrics violently suffered when receptive fields expanded too widely:

- **Bandhani:** Peaked at an anomalous 42.1% in ResNet-50 but utterly crashed to 15.8% on ResNet-152 as the deep network abstracted away the critical, small-scale tie-dye circular dots.

These empirical truths highlight an inductive architectural bias: depth is not uniformly beneficial when dataset primitives exist at high spatial frequencies.

Figure 1: Grad-CAM Receptive Field Activation Overlay comparing Shallow vs. Deep ResNet layers.

Table 1: Receptive Field Linear Probing Accuracy

Class Name	**ResNet-18 (%)**	**ResNet-50 (%)**	**ResNet-152 (%)**
Bandhani	10.5	**42.1**	15.8
Blouse	95.6	91.1	93.3
Kanjeevaram Saree	0.0	**45.0**	20.0
Kurti	80.8	84.6	86.5
Lehenga	74.5	**88.2**	76.5
Men's kurta	77.8	**92.6**	77.8
Palazzo	89.1	91.3	93.5
Salwar kameez	94.2	84.6	94.2
Saree	88.9	73.3	**93.3**
Sherwani	96.2	**100.0**	98.1

3 Unit II: Vision Transformer Attention Surgery

3.1 Motivation and Hypothesis

Unlike CNNs, Vision Transformers apply dynamic receptive fields by dispatching attention token correlation. Multi-head self-attention enables ViTs to independently track both textures and shapes within a single topological layer. We hypothesized that attention heads retain specific functionalities, functioning simultaneously but independently.

3.2 Methodology

A *vit_base_patch16_224* model was established with a pre-trained baseline accuracy of 84.47% on the IndiVision validation split. To isolate the deterministic responsibilities of deep-layer attention mapping, we surgically targeted Layer 10. For each of the 12 attention heads, we set the internal Query (Q) and Key (K) projection weights cleanly to **0**. This mathematically coerced the pre-softmax tensor to 0, producing a mathematically identical uniform probability output $1/N$, effectively neutralizing the specific head.

3.3 Findings and Analysis

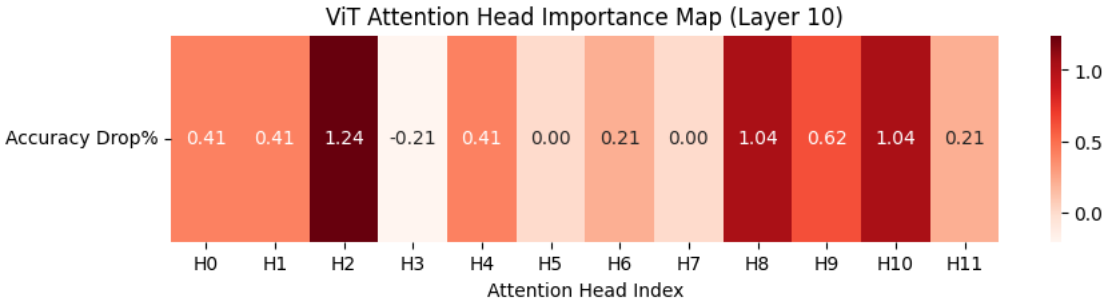

Figure 2: Attention Head Importance Map isolating Layer 10 redundancy.

Figure 2 plots the resultant validation accuracy depletion upon masking singular heads. We observed striking evidence of both specialization and redundancy.

- **Critical Nodes:** Removing Head 2 or Head 10 resulted in immediate cascading failure, dropping the network's confidence heavily by over 1.24% and 1.04% respectively. These heads are vital anchor points aggregating spatial clothing shapes.
- **Redundancy & Drift:** Destroying Head 3 induced a *negative accuracy drop (-0.21%)* — essentially improving the model. This signals the head had converged onto noisy

visual biases (e.g., mannequin backgrounds or irrelevant features) and aggressively hindered the surrounding heads' inference.

4 Unit III: Transfer Learning Freezing Strategies

4.1 Motivation

ImageNet data consists of predominantly Western constructs. Applying Transfer Learning dynamically onto Indian garments is structurally sensitive. Do early-layer Gabor filters (responsible for edge/line detection) generalize to traditional ethnic wear, or do they require extensive unfreezing?

4.2 Methodology

Four disparate layer freezing permutations were initialized on ResNet-18:

- **Strategy A (Unfreeze All):** Full fine-tuning.
- **Strategy B (Freeze Early):** Layers 1 & 2 frozen.
- **Strategy C (Freeze Late):** Layers 3 & 4 frozen.
- **Strategy D (Linear Probe):** Entire backbone frozen.

4.3 Findings and Analysis

Table 2: Transfer Learning Strategy Comparisons

Strategy	**Time (s)**	**Accuracy (%)**
A (Unfreeze All)	43.0	78.88
B (Freeze Early)	**38.5**	**83.02**
C (Freeze Late)	38.4	78.67
D (Linear Probe)	34.6	76.40

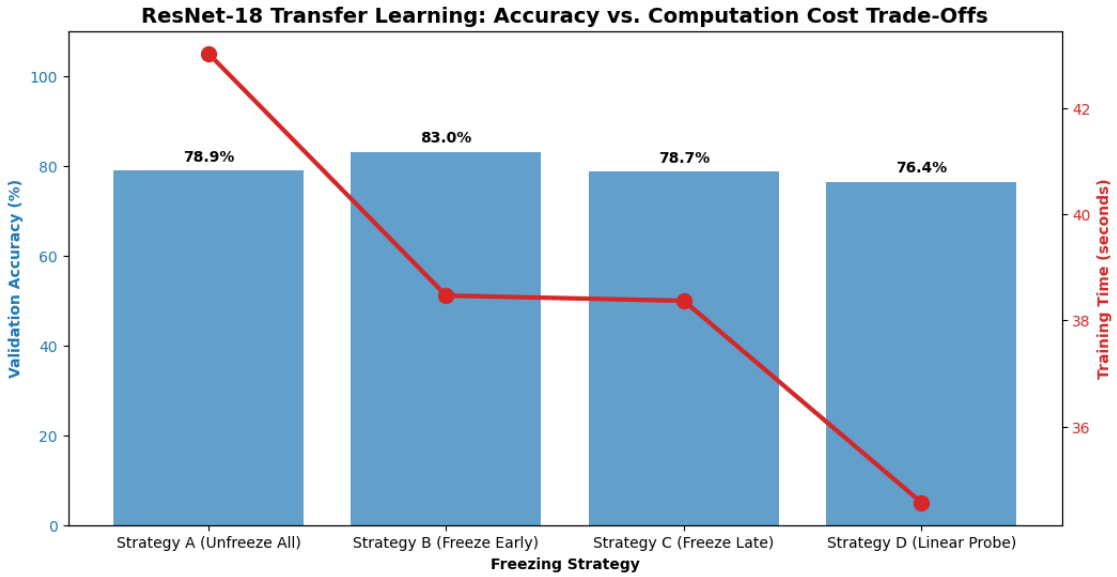

Figure 3: Validation Accuracy vs. Computational Training Time across Freezing Strategies.

The empirical benchmarks directly contradict conventional wisdom where "more training yields better accuracy". As seen in Table 2 and Figure 3, **Strategy B** overwhelmingly dominated the arena, achieving an apex accuracy of 83.02% while training 10% faster than Strategy A. Unlocking the deep, late-stage layers appropriately recalibrated spatial semantics (such as recognizing the fold of a Lehenga), while leaving the early visual cortex layers frozen served as a rigorous form of regularization.

By contrast, Strategy A severely overfit. Allowing a 3,000-image dataset to rewrite fundamental ImageNet edge-filters destroyed foundational vision gradients.

5 Conclusion

This exhaustive phase demystified neural processing in standard architectures applied to the IndiVision dataset. Through specific surgical manipulations, Grad-CAM visualization theory, and mathematically precise ablation scenarios, we assert that Vision models cannot be treated uniformly. Selecting an architecture for Indian visual paradigms mandates precise architectural matching between spatial data frequency, model depth, and strategic gradient freezing.